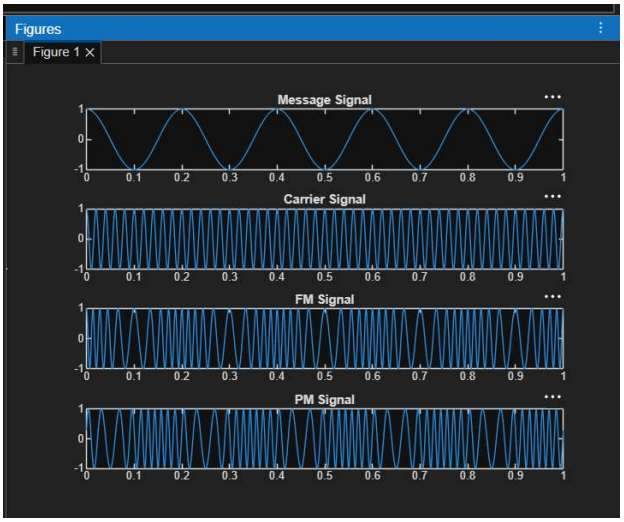
ECA0801 - ANALOG AND DIGITAL COMMUNICATION

Exp 6: GENERATION AND ANALYSIS OF FREQUENCY MODULATED (FM) AND OF PHASE MODULATED (PM) SIGNALS USING MATLAB.

Aim:

To generate and analyze Frequency Modulated (FM) and Phase Modulated (PM) signals using MATLAB by varying the instantaneous frequency and phase of a carrier signal in accordance with the message signal.

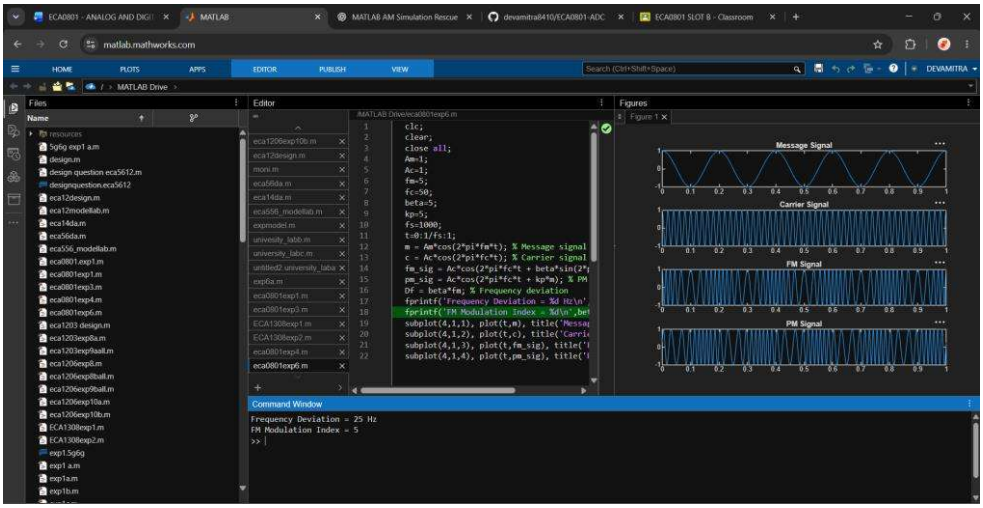


Program:

```
clc;
clear;
close all;
Am=1;
Ac=1;
fm=5;
fc=50;
beta=5;
kp=5;
fs=1000;
t=0:1/fs:1;
m = Am*cos(2*pi*fm*t); % Message signal
c = Ac*cos(2*pi*fc*t); % Carrier signal
fm_sig = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t)); % FM
pm_sig = Ac*cos(2*pi*fc*t + kp*m); % PM
Df = beta*fm; % Frequency deviation
```

```
fprintf('Frequency Deviation = %d Hz\n',Df);
fprintf('FM Modulation Index = %d\n',beta);
subplot(4,1,1), plot(t,m), title('Message Signal')
subplot(4,1,2), plot(t,c), title('Carrier Signal')
subplot(4,1,3), plot(t,fm_sig), title('FM Signal')
subplot(4,1,4), plot(t,pm_sig), title('PM Signal')
```

Output:



QUESTION 1

A community FM radio station is commissioning a new FM transmitter for broadcasting educational programs. Before deployment, the broadcast engineer must verify that the transmitter produces the desired modulation index to ensure efficient bandwidth utilization and high-quality audio transmission. The transmitter exhibits a maximum frequency deviation of 3 kHz when modulated by an audio signal of 1 kHz. The engineer is required to validate the theoretical modulation index and compare it with the MATLAB simulation results. Given: Carrier Amplitude, Ac = 10V, Carrier Frequency, fc = 20kHz, Message Amplitude, Am = 5V, Message Frequency, fm = 1kHz, Maximum Frequency Deviation, Δf = 3kHz, Sampling Frequency, fs = 200kHz

Tasks:

1. Calculate the theoretical modulation index (β)of the FM signal.

2. Develop a MATLAB program to generate the message signal and carrier signal.

3. Generate the FM signal using the given parameters.

4. Display the calculated modulation index in the MATLAB Command Window.

5. Plot the message signal, carrier signal, and FM signal.

6. Compare the theoretical modulation index with the value used in the MATLAB simulation and comment on the result.

Program:

clc; clear; close all;

% Parameters Ac = 10; % Carrier amplitude (V) fc = 20e3; % Carrier frequency (Hz) Am = 5; % Message amplitude (V) fm = 1e3; % Message frequency (Hz) fs = 200e3; % Sampling frequency (Hz) df = 3e3; % Frequency deviation (Hz)

% Calculate modulation index beta = df/fm; fprintf('FM Modulation Index = %.2f\n', beta);

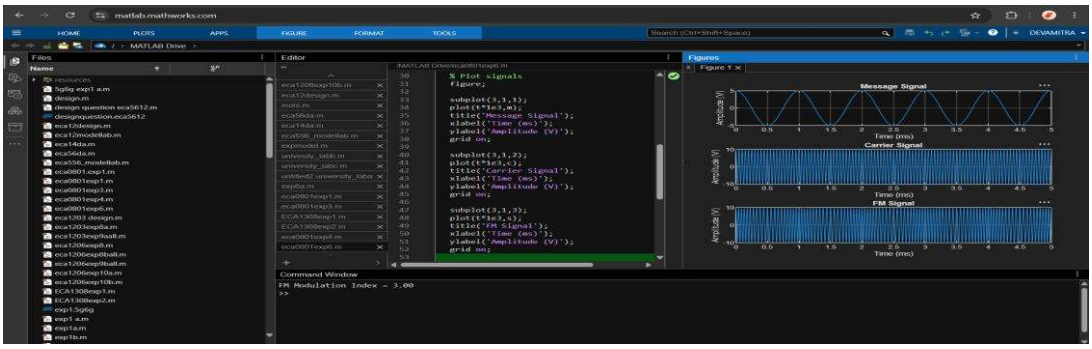
% Time vector t = 0:1/fs:5e-3;

% Message signal m = Am*cos(2*pi*fm*t);

% Carrier signal c = Ac*cos(2*pi*fc*t);

% FM signal s = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));% Plot signals figure;subplot(3,1,1); plot(t*1e3,m); title('Message Signal'); xlabel('Time (ms)'); ylabel('Amplitude (V)'); grid on;subplot(3,1,2); plot(t*1e3,c); title('Carrier Signal'); xlabel('Time (ms)'); ylabel('Amplitude (V)'); grid on;subplot(3,1,3); plot(t*1e3,s); title('FM Signal'); xlabel('Time (ms)'); ylabel('Amplitude (V)'); grid on;

Output:



QUESTION 2

A satellite communication company is testing a Phase Modulation (PM) transmitter for telemetry applications. To ensure reliable data transmission, the engineer must verify the phase deviation produced by the transmitter and observe its effect on the transmitted waveform. The system is designed to produce a maximum phase deviation of 2 radians when modulated by a 500 Hz message signal. Given: Carrier Amplitude, Ac = 10V, Carrier Frequency, fc = 15kHz, Message Amplitude, Am = 4V, Message Frequency, fm = 500Hz, Phase Deviation Constant, kp = 2rad/V, Sampling Frequency, fs = 150kHz

Tasks:

1. Calculate the maximum phase deviation of the PM signal.
2. Develop a MATLAB program to generate the message signal and carrier signal.
3. Generate the PM signal using the given parameters.
4. Display the calculated phase deviation in the MATLAB Command Window.
5. Plot the message signal, carrier signal, and PM signal.
6. Compare the characteristics of the PM signal with the original message signal and comment on the effect of phase modulation.

Program:

```
clc; clear; close all;

% Parameters Ac = 10; % Carrier amplitude (V) fc = 15e3; % Carrier frequency (Hz) Am = 4; % Message amplitude (V) fm = 500; % Message frequency (Hz) kp = 2; % Phase deviation constant (rad/V) fs = 150e3; % Sampling frequency (Hz)

% Calculate phase deviation phase_dev = kp * Am; fprintf('Maximum Phase Deviation = %.2f radians\n', phase_dev);

% Time vector t = 0:1/fs:6e-3;

% Message signal m = Amcos(2pifmt);

% Carrier signal c = Accos(2pifct);
```

% PM signal PM = Ac*cos(2*pi*fc*t + kp*m);

% Plot signals figure;

subplot(3,1,1); plot(t*1e3,m); title('Message Signal'); xlabel('Time (ms)'); ylabel('Amplitude (V)'); grid on;

subplot(3,1,2); plot(t*1e3,c); title('Carrier Signal'); xlabel('Time (ms)'); ylabel('Amplitude (V)'); grid on;

subplot(3,1,3); plot(t*1e3,PM); title('Phase Modulated Signal'); xlabel('Time (ms)'); ylabel('Amplitude (V)'); grid on;

Output:

